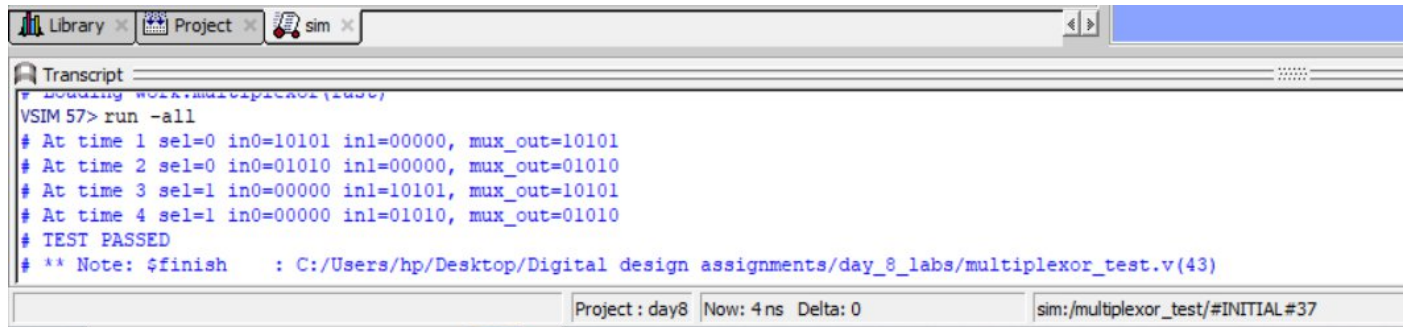


Multiplexor:

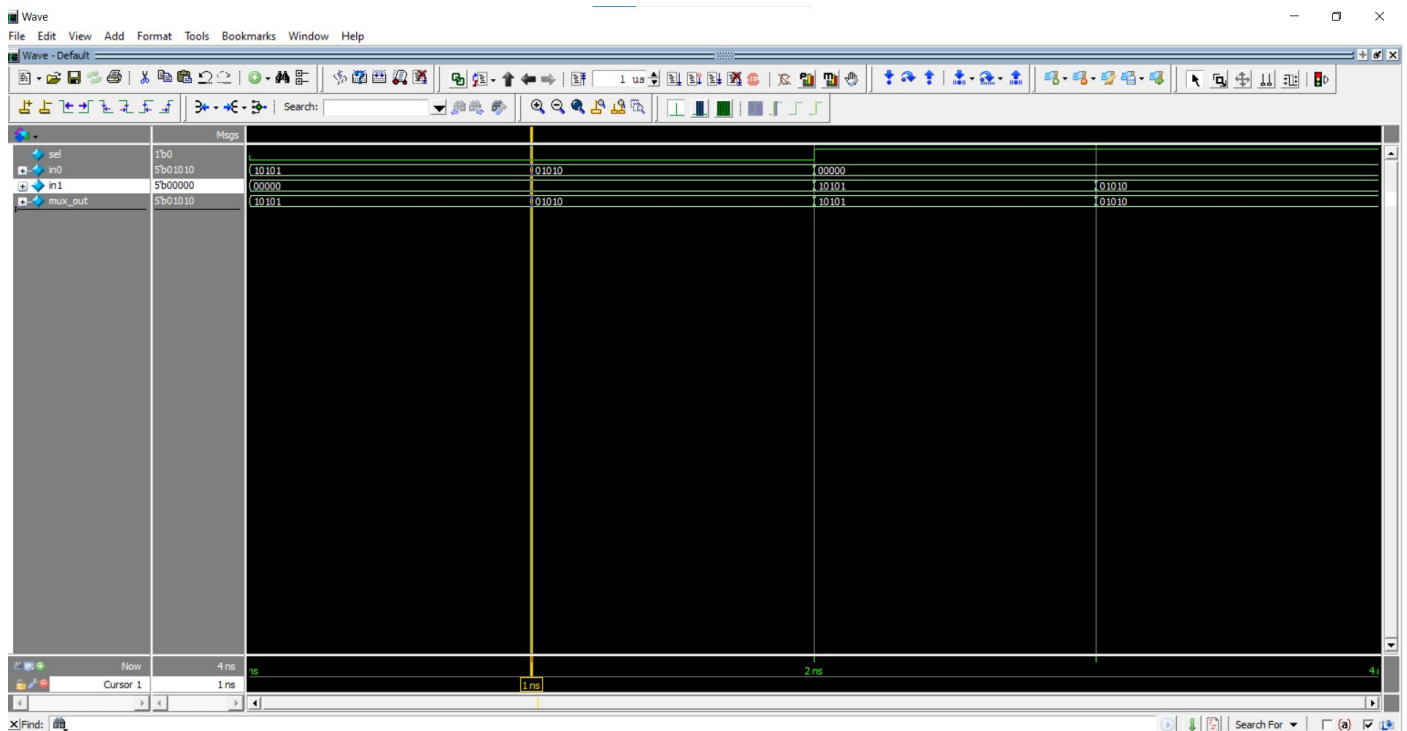
Result:

All test cases produced outputs matching the expected values:



The screenshot shows a simulation transcript window with the following text:

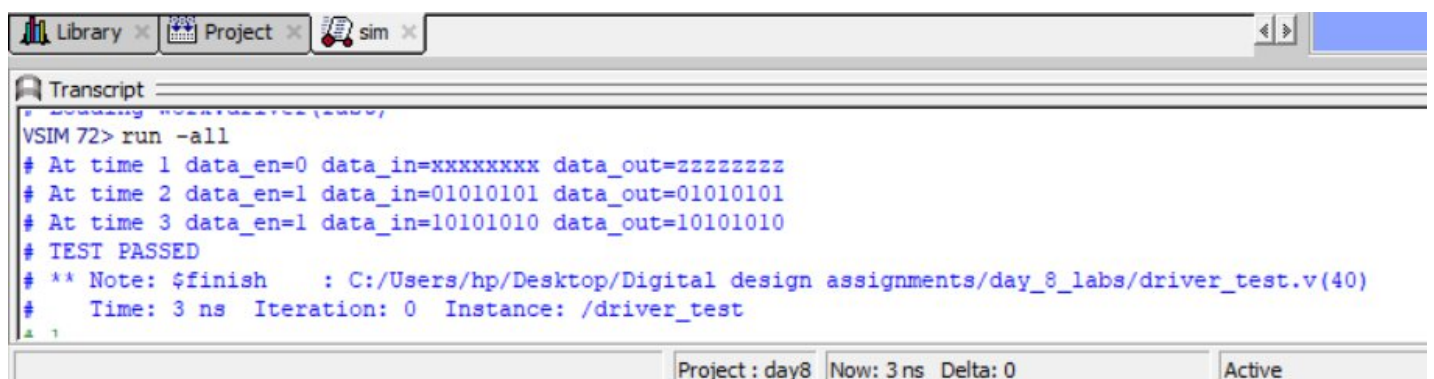
```
Library x Project x sim x
Transcript
# Loading work/multiplexor_test.v
VSIM 57> run -all
# At time 1 sel=0 in0=10101 in1=00000, mux_out=10101
# At time 2 sel=0 in0=01010 in1=00000, mux_out=01010
# At time 3 sel=1 in0=00000 in1=10101, mux_out=10101
# At time 4 sel=1 in0=00000 in1=01010, mux_out=01010
# TEST PASSED
# ** Note: $finish      : C:/Users/hp/Desktop/Digital design assignments/day_8_labs/multiplexor_test.v(43)
Project: day8 Now: 4 ns Delta: 0 sim:/multiplexor_test/#INITIAL#37
```



Driver:

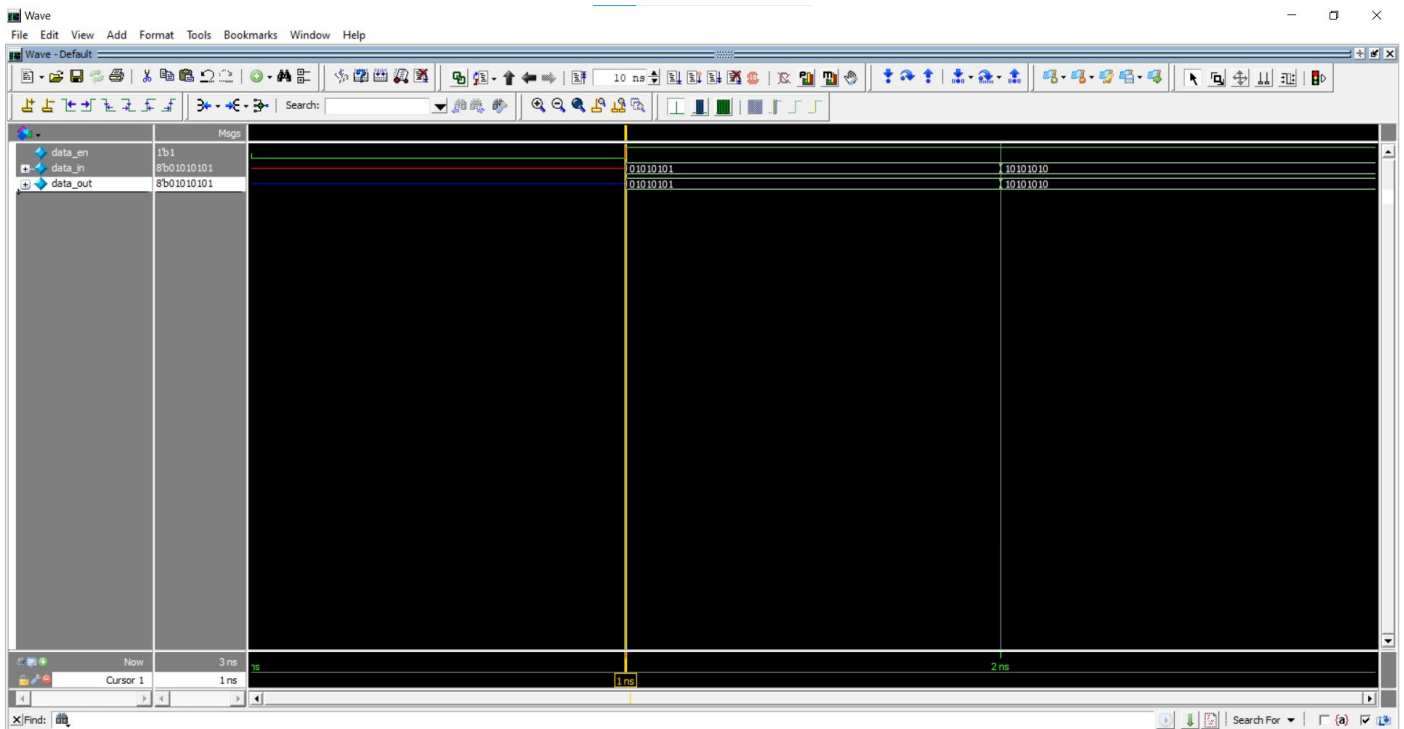
Result:

All test cases produced outputs matching the expected values:



The screenshot shows a simulation transcript window with the following text:

```
Library x Project x sim x
Transcript
# Loading work/driver_test.v
VSIM 72> run -all
# At time 1 data_en=0 data_in=xxxxxxx data_out=zzzzzzzz
# At time 2 data_en=1 data_in=01010101 data_out=01010101
# At time 3 data_en=1 data_in=10101010 data_out=10101010
# TEST PASSED
# ** Note: $finish      : C:/Users/hp/Desktop/Digital design assignments/day_8_labs/driver_test.v(40)
# Time: 3 ns Iteration: 0 Instance: /driver_test
Project: day8 Now: 3 ns Delta: 0 Active
```



Alu :

Result:

All test cases produced outputs matching the expected values:

```
Library Project sim
Transcript
add wave sim:/alu_test/*
VSIM 87> run -all
# At time 1 opcode=000 in_a=01000010 in_b=10000110 a_is_zero=0 alu_out=01000010
# At time 2 opcode=001 in_a=01000010 in_b=10000110 a_is_zero=0 alu_out=01000010
# At time 3 opcode=010 in_a=01000010 in_b=10000110 a_is_zero=0 alu_out=11001000
# At time 4 opcode=011 in_a=01000010 in_b=10000110 a_is_zero=0 alu_out=00000010
# At time 5 opcode=100 in_a=01000010 in_b=10000110 a_is_zero=0 alu_out=11000100
# At time 6 opcode=101 in_a=01000010 in_b=10000110 a_is_zero=0 alu_out=10000110
# At time 7 opcode=110 in_a=01000010 in_b=10000110 a_is_zero=0 alu_out=01000010
# At time 8 opcode=111 in_a=01000010 in_b=10000110 a_is_zero=0 alu_out=01000010
# At time 9 opcode=111 in_a=00000000 in_b=10000110 a_is_zero=1 alu_out=00000000
# TEST PASSED
# ** Note: $finish      : C:/Users/hp/Desktop/Digital design assignments/day_8_labs/alu_test.v(57)
Project : day8 Now: 9 ns Delta: 0 sim:/alu_test/#INITIAL #46
```

